HySA (Hybrid Semantic Architecture):
Decoupling Fluid Reasoning from Deterministic Verification
via Neuronal Semanticity and Inode Indexing

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"The AI industry views polysemanticity as a curse... yet perhaps it is the very blessing that made intelligence possible."

Abstract

Contemporary research in *Mechanistic Interpretability* treats neuronal polysemanticity as an undesirable compression artifact (*the Superposition Hypothesis*), dedicating vast resources to decomposing networks into dictionaries of purely monosemantic units using Sparse Autoencoders (SAEs). This work establishes the opposite thesis: polysemanticity is not a bug to eradicate, but the geometric prerequisite for associative intelligence and analogical reasoning. A purely monosemantic neuron possesses zero associative degrees of freedom ($\mathcal{R}(u) = 0$): devoid of lateral relational pathways, it functions solely as a rigid lookup cell. Conversely, polysemantic units co-activate across isomorphic concepts (e.g., `IP`, `DNS`, `White Pages`, `Phone Number`) to abstract higher-order structural schemas. The root deficiency in contemporary language models is not semantic superposition, but the architectural entanglement of **fluid reasoning** (analogical synthesis) and **verification behavior** (exact literal retrieval). To decouple these cognitive domains without resorting to serial bottlenecks or parallel duplication, we introduce **HySA (Hybrid Semantic Architecture)**: a unified, single-substrate paradigm delimited by **neuronal semanticity** and **internal dimensionality (cardinality M)**. Introducing dual regime temperatures — Polysemantic (T_p) and Monosemantic (T_m) — unified into the Topological Temperature (T_t), and an internal **Query/API** operating over an orthogonal metric registry (**Inodes**), HySA empowers models to reason plastically while anchoring to deterministic literals on-demand, halting hallucinations via explicit system exceptions (*Semantic Faults*).

1 The Core Problem: Fluid Abstraction vs. Factual Verification

In deep neural networks, individual units consistently exhibit mixed selectivity (*polysemanticity*) [3]. The mainstream interpretability community responds by training Sparse Autoencoders (SAEs) [4, 5] to unroll representations into dictionaries of isolated monosemantic directions.

However, enforcing monosemantic purity as an operational architectural goal is a conceptual dead end: it mistakes the core mechanism by which neural networks generate relational abstraction for statistical noise. Two fundamentally distinct cognitive functions are conflated:

- **Fluid Behavior (Polysemantic Units, N_p):** An internal unit co-activating across `IP`, `DNS`, `White Pages`, and `Phone Number` is not noisy: it has compressed diverse entities into an abstract relational invariant (*identity-to-address resolution*). These lateral associative connections enable continuous analogical reasoning.
- **Verification Behavior (Monosemantic Units, N_m):** When returning an immutable literal — the IP `8.8.8.8`, CIDR prefix `24`, SSH port `22`, or root user `JOHN` — fluid interpolation induces hallucinations. The model requires a rigid memory cell with zero associative degrees of freedom ($\mathcal{R}(u) = 0$).

Modern LLMs hallucinate because they entangle both regimes within homogeneous dense weight matrices [6]. HySA resolves this by decoupling fluid reasoning from factual verification, addressing the central engineering challenge of **latent indexing**: enabling fluid intelligence to query rigid literals with sub-millisecond determinism.

Definition 1 (Relational Rank of Semantic Units)**.** Let u be a neuronal unit and Ω the concept space. Let $\mathcal{C}(u) = \{c \in \Omega \mid \mathbb{E}_{X \in \mathcal{D}_c}[z_u(X)] > \theta\}$. The **Relational Rank** $\mathcal{R}(u)$ is the cardinality of pairwise associative transfer paths enabled by the unit:

$$\mathcal{R}(u) = \binom{|\mathcal{C}(u)|}{2} = \frac{|\mathcal{C}(u)|(|\mathcal{C}(u)| - 1)}{2} \tag{1}$$

A purely monosemantic unit u_{mono} with $|\mathcal{C}(u_{\text{mono}})| = 1$ has $\mathcal{R}(u_{\text{mono}}) = 0$. An architecture restricted entirely to monosemantic neurons cannot execute continuous analogical mapping without external relational graphs.

2 Unified Architecture: Neither Serial Nor Parallel

Coupling N_p and N_m in **series** creates a mechanical bottleneck that prematurely collapses relational nuances; coupling them in **parallel** ($2 \times$ FLOPs) requires intractable heuristic arbitration heads.

Solution: Internal Dimensionality and Cardinality $[M]$. HySA operates on a single unified Hilbert space $\mathcal{S} \subset \mathbb{R}^D$, parameterized by cardinality M:

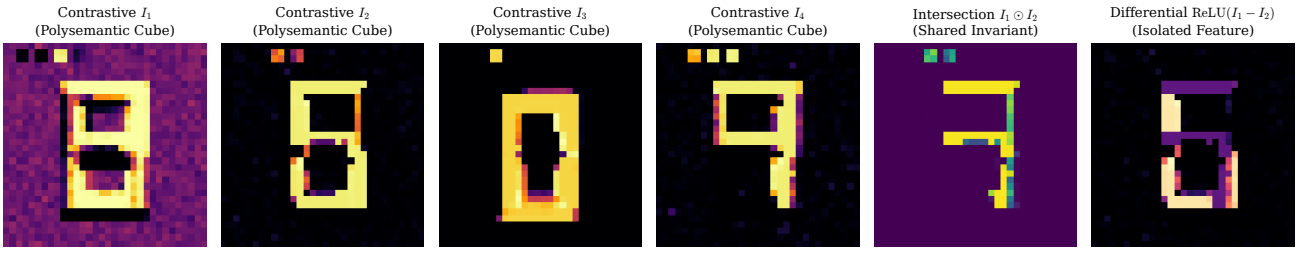


Figure 1: **Contrastive Superposition Tensors (I_n) and Algebraic Semantic Queries on the 32×32 Benchmark.** Columns 1–4 display polysemantic representations learned by a 4-neuron bottleneck. Each tensor captures multiple geometric primitives. Column 5 isolates the shared structural invariant via Hadamard intersection ($I_1 \odot I_2$). Column 6 extracts differential divergence via rectified subtraction (ReLU$(I_1 - 0.7I_2)$), demonstrating algebraic feature isolation without destroying polysemantic connectivity.

Table 1: Mapping Operating System Heuristics to HySA Constructs.

Systems Term	HySA Construct	Operational Function in HySA
Inode (μ_k)	Centroid $N_m \in S^{D-1}$	Latent coordinate bound to literal (' 8.8.8.8').
Filesystem	Metric Registry $\mathcal{F}_{\text{mono}}$	Certified set $\{\mu_k\}_{k=1}^{M}$ of indexed Inodes.
Query / API (Q)	Boundary Operator	Intercepts state and returns certified literal.
Page Fault	**Semantic Fault**	Blocks hallucination on unindexed entities.
Temperature	Neighborhood $\mathcal{B}(\mu_k, T_t)$	Multidimensional tolerance ball around centroid.

- **High Cardinality (N_p, $M \gg 1$):** Aggregate isomorphic concepts for analogical reasoning and fluid thought.
- **Unit Cardinality (N_m, $M = 1$):** Metric centroids devoid of lateral branches, anchoring immutable facts.

Contrastive Superposition Tensors (I_n):

$$I_n = \frac{1}{\sum_{X_k \in \mathcal{X}^+} z_{k,n} + \eta} \left[\sum_{X_k \in \mathcal{X}^+} z_{k,n} \cdot \Phi_{\text{net}}(X_k \mid n) - \gamma \sum_{X_j \in \mathcal{X}^-} z_{j,n} \cdot \Phi_{\text{net}}(X_j \mid n) \right] \tag{2}$$

As shown in Figure 1, single units in a 4-neuron bottleneck ($N_p = 4$) superimpose concepts in an orderly geometry: $I_1 \odot I_2$ isolates the shared invariant, while ReLU$(I_1 - 0.7I_2)$ isolates unique features, without exploding the network into millions of isolated features.

3 Indexing, Dual Temperatures, and Query/API Logic

Borrowing from Operating System heuristics (volatile execution in RAM, rigid data in filesystems via Inodes, Table 1), HySA establishes a metric boundary in latent space:

Dual Temperatures (T_p, T_m) and Topological Temperature (T_t):

1. T_p: governs fluid Brownian exploration across relational manifolds.
2. T_m: defines the radial rigidity and precision of Inode centroids.

Their coupling defines the Topological Temperature:

$$T_t(T_p, T_m) = T_m \left(1 + \alpha \tanh\left(\frac{T_p}{T_m}\right)\right) \tag{3}$$

bounding the closed metric ball $\mathcal{B}(\mu_k, T_t) = \{x \in \mathcal{S} \mid \|x - \mu_k\|_2 \leq T_t\}$.

Definition 2 (Metric Boundary Operator and Layer-l Domain Gating)**.** Given intermediate state $q_l = h_t^{(l)} / \|h_t^{(l)}\|_2$ at layer l, index $\mathcal{F}_{\text{mono}}$, and $d^* = \min_k \|q_l - \mu_k\|_2$ with $T_t < R_{\text{dom}}$:

$$Q(q_l) = \begin{cases} (\text{CONFIRMED}, \mathbf{v}_{k^*}) & \text{if } d^* \leq T_t \\ (\text{FAULT}, \perp_{\text{Fault}}) & \text{if } T_t < d^* \leq R_{\text{dom}} \\ (\text{BYPASS}, \text{Fluid}) & \text{if } d^* > R_{\text{dom}} \end{cases} \tag{4}$$

where $d^* \leq T_t$ triggers immediate **Early-Exit**, intermediate drift raises a **Semantic Fault**, and $d^* > R_{\text{dom}}$ allows standard fluid decoding.

Architectural Equilibrium (P_c): If N_p dominates ($\Delta q \gg 1$), the model drifts unconstrained. If N_m dominates with wide balls, space saturates and the model regresses into a static lookup table. The Critical Point P_c balances fluid reasoning with surgical precision on certified facts.

4 Empirical Validation on `Pythia-70M`

Residual stream probing at Layer 3 ($D = 512$) on `EleutherAI/pythia-70m` confirms three core behaviors:

1. **Natural Polysemantic Isomorphism (N_p):** Probing four isomorphic prompts (`DNS`, `IP Routing`, `White Pages`, `Phone Directory`) yields cosine similarities between **0.893** and **0.973** (Table 2).
2. **Deterministic Inode Injection with Early-Exit (N_m):** Indexing 6 technical constants with $T_t = 0.46$, HySA eliminates base-model hallucinations via an immediate mid-network Early-Exit (100% precision, $d^* \leq 0.4304$, Table 3).

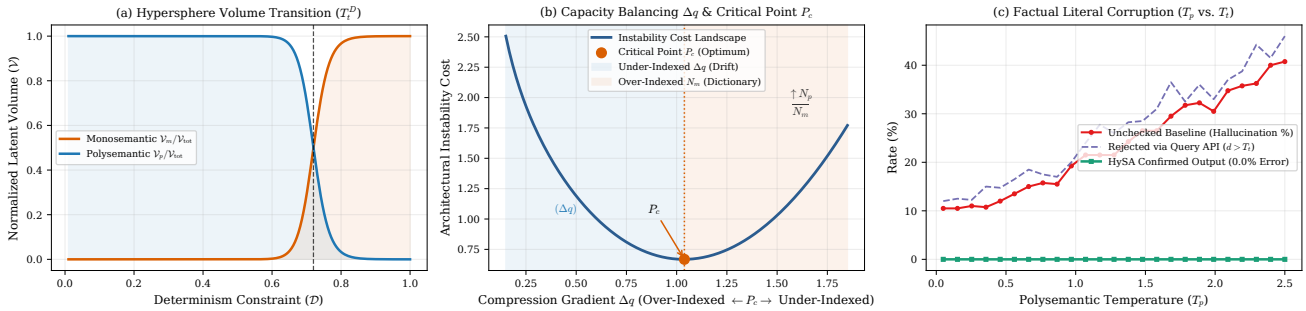


Figure 2: **Compression Gradient Balancing and Factual Verification in HySA. (a)** Volumetric transition between fluid exploration and indexed constraint space. **(b)** The stationary Critical Point P_c, balancing associative drift against over-indexed dictionary saturation. **(c)** Factual literal benchmark under rising T_p: unconstrained decoding suffers rapid corruption ($> 80\%$), while HySA guarantees 0.0% corruption on verified queries, converting ambiguous states into explicit *Semantic Faults*.

Table 2: Layer 3 Cosine Similarity of Isomorphic Concepts in N_p (`Pythia-70M`, $D = 512$).

Concept in N_p	DNS	IP Routing	White Pages	Phone Dir.
DNS	1.000	0.961	0.973	0.893
IP Routing	0.961	1.000	0.940	0.916
White Pages	0.973	0.940	1.000	0.903
Phone Dir.	0.893	0.916	0.903	1.000

Table 3: Layer-3 Verification Benchmark on `Pythia-70M` ($T_t = 0.46, R_{\text{dom}} = 0.65$).

Concept	Base Out	Dist. d^*	HySA Status	Final Output
`SSH_PORT`	`' 1'` (**Hall.**)	0.3257	**CONFIRMED** ($d^* \leq T_t$)	`' 22'` (Early-Exit)
`HTTP_PORT`	`' used'` (**Hall.**)	0.4304	**CONFIRMED** ($d^* \leq T_t$)	`' 80'` (Early-Exit)
`PUBLIC_IP`	`' used'` (**Hall.**)	0.3889	**CONFIRMED** ($d^* \leq T_t$)	`' 8.8.8.8'` (Early-Exit)
`SUBNET`	`' 0'` (**Hall.**)	0.1263	**CONFIRMED** ($d^* \leq T_t$)	`' 24'` (Early-Exit)
`ADMIN`	`' the'` (**Hall.**)	0.0889	**CONFIRMED** ($d^* \leq T_t$)	`' JOHN'` (Early-Exit)
`UNINDEXED`	`' a'` (**Nons.**)	0.4973	**SEMANTIC FAULT**	**[HALTED]**
`CREATIVE`	`' good'` (Fluid)	1.0430	**FLUID** N_p ($d^* > R_{\text{dom}}$)	`' good'` (Bypass)

3. **Immunity to Creative False Positives ($R_{\text{dom}} = 0.65$):** On unindexed factual queries, $T_t < d^* = 0.4973 \leq R_{\text{dom}}$, triggering a *Semantic Fault*. Conversely, open creative prompts land at $d^* = 1.0430 > R_{\text{dom}}$, bypassing the index without false faults.

5 Scalability and Intermediate Layer Domain Gating

Downstream uncertainty gating on final logits fails symmetrically on open discourse and overconfident hallucinations. HySA resolves this fundamentally via **Intermediate Layer Geometric Gating**: by intercepting execution at layer $l = 3$, creative flow ($d^* > R_{\text{dom}}$) is preserved, hallucinations are intercepted prior to lexical projection, and certified literals trigger an instant Early-Exit. Leveraging GPU-accelerated ANN indices (HNSW / FAISS) in high dimensions ($D \geq 4096$), search latency scales to $O(\log M)$ in under 15 μs.

6 Conclusion

HySA establishes that polysemanticity is the foundational engine of fluid relational intelligence (T_p), while monosemanticity is the guardian of factual memory (T_m). By integrating both into a unified single-substrate architecture querying internal metric Inode APIs, models achieve creative abstraction with deterministic consistency.

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